

Population-level prediction of depression, Alzheimer’s disease, and Parkinson’s disease incidence using artificial intelligence: implications for health service planning

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Background and aims · Forecast weekly neuropsychiatric notifications to support planning.

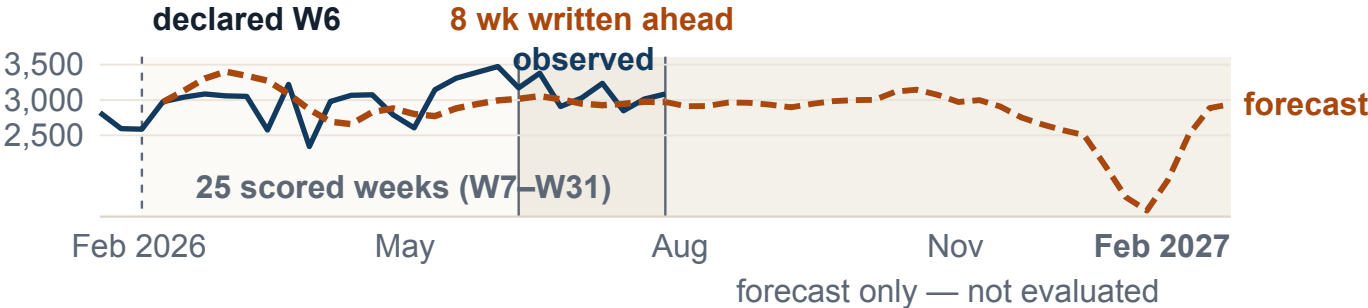
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Depression: 74,675 forecast vs 75,859 observed over 25 weeks (−1.6%).

National sMAPE: 8.7% depression · 10.6% Parkinson · 14.4% Alzheimer Forecast error (sMAPE): 0% is exact; lower is better.

Depression · 52-week forecast and observed notifications

National counts · 9 Feb 2026–1 Feb 2027



Methods · Retrospective study

SINAVE = Mexico’s National Epidemiological Surveillance System; Dec 2013–Jan 2026.

333 series; expanding-window cross-validation. One forecast selected per series.

Prophet · DeepAR · Ensemble · Stacking

Evaluation: 297 series (national + 32 states, by sex); four regions excluded.

File: 5 Jun 2026. Only W24–W31 were written ahead; W7–W23 had occurred.

Results · Updated evaluation

|            | 25 wk | 8 wk  | Observed → forecast      |
|------------|-------|-------|--------------------------|
| Depression | 8.7%  | 5.0%  | 75,859 → 74,675<br>−1.6% |
| Parkinson  | 10.6% | 6.7%  | 3,832 → 3,748<br>−2.2%   |
| Alzheimer  | 14.4% | 17.8% | 1,100 → 1,046<br>−4.9%   |

State-level variation · sMAPE

Depression · W7–W31 · 25 wk



National 8.7%  
Median 21.6%  
Tlaxcala 46.7%

Conclusions

- Supports planning at national level.
- State-level errors remain larger; national results cannot stand in for local performance.

Limitations

DeepAR training cut-off: declared, not artifact-verified.  
Notifications ≠ incidence; patient outcomes untested.



METHODS, DATA  
AND CODE

EDIT Equality: sex-stratified  
Development: age/life stage not assessed · Inclusion: open public data  
Transcultural: Mexico  
epiforecast.mx/wcp2026.html

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